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Laboratory 8: Wired Communication

I. Objectives

- To understand the basic operation of UART, I²C, and SPI communication.
- To test wired communication using an ESP32.
- To observe how jumper connections affect communication results.

II. Materials

- ESP32 development board
- USB cable
- Computer with Arduino IDE
- OLED display (I²C)
- Jumper wires

III. Procedure

A. UART Loopback Test

1. Upload the UART loopback program to the ESP32.
2. Connect GPIO17 (TX) to GPIO16 (RX) using a jumper wire.
3. Open the Serial Monitor and observe the transmitted and received data.
4. Remove the jumper wire and observe the result.

B. I²C Scanner Test

1. Connect the OLED display to the ESP32.
2. Connect SDA to GPIO21 and SCL to GPIO22.
3. Upload and run the I²C scanner program.
4. Observe the detected I²C device address.
5. Disconnect the OLED and run the scanner again.

C. SPI Loopback Test

1. Upload the SPI loopback program to the ESP32.
2. Connect MOSI (GPIO23) to MISO (GPIO19) using a jumper wire.
3. Observe the transmitted and received data in the Serial Monitor.
4. Remove the jumper wire and observe the result.

IV. Results and Observations

Test	Connection	Observation
UART	Jumper connected	Transmitted PING was received successfully.
UART	Jumper removed	Only the transmitted message appeared; no echo was received.
I²C	OLED connected	The scanner detected the OLED's I²C address.
I²C	OLED disconnected	No I²C devices were found.
SPI	MOSI connected to MISO	Received 0xA5; loopback test passed.
SPI	Jumper removed	Received 0x00; loopback test failed as expected.

V. Guide Questions

1. Why does UART require the same baud rate on both devices, while SPI uses a clock signal?

UART requires the same baud rate so the transmitter and receiver can interpret data at the same speed. SPI uses a clock signal to synchronize data transmission.

2. What happens if two I²C devices have the same address?

Two devices with the same address may cause communication conflicts because the controller cannot reliably distinguish between them.

3. What is the purpose of the Chip Select (CS) pin in SPI communication?

The CS pin selects the SPI device to communicate with. It is usually LOW when the device is selected and HIGH when it is deselected.

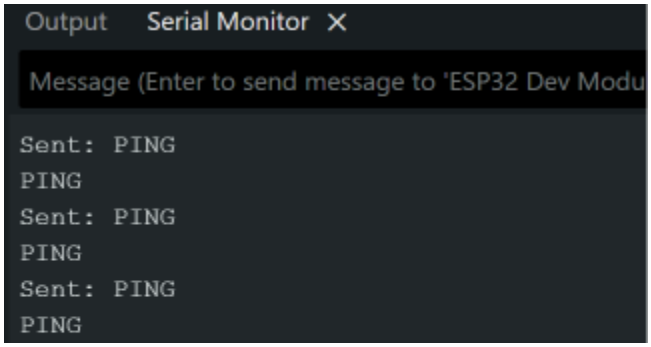
VI. Conclusion

The laboratory activity demonstrated UART, I²C, and SPI communication using the ESP32. The tests showed that correct wiring is important for successful communication and that removing the required connections affects the results.

VII. Documentation

1. UART Loopback Test

Figure 1.1. Successful UART Loopback Test



Observation: The transmitted PING message was received successfully.

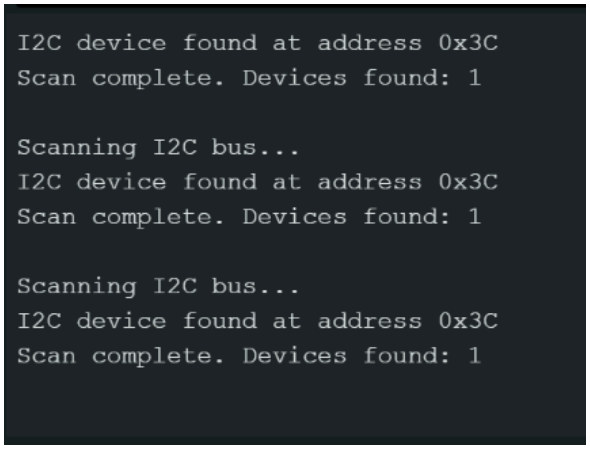
Figure 1.2. Failed UART Loopback Test



Observation: No echo was received because the TX and RX pins were disconnected.

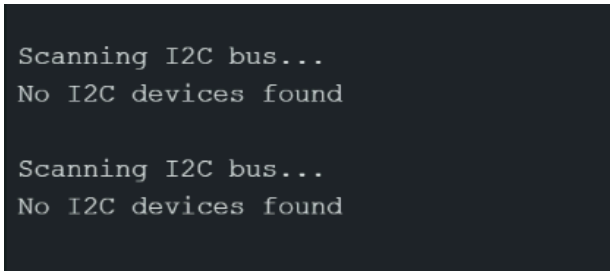
2. I²C Scanner Test — OLED Display

Figure 2.1. Successful I²C Scanner Test



Observation: The scanner detected the OLED display through the SDA and SCL connections.

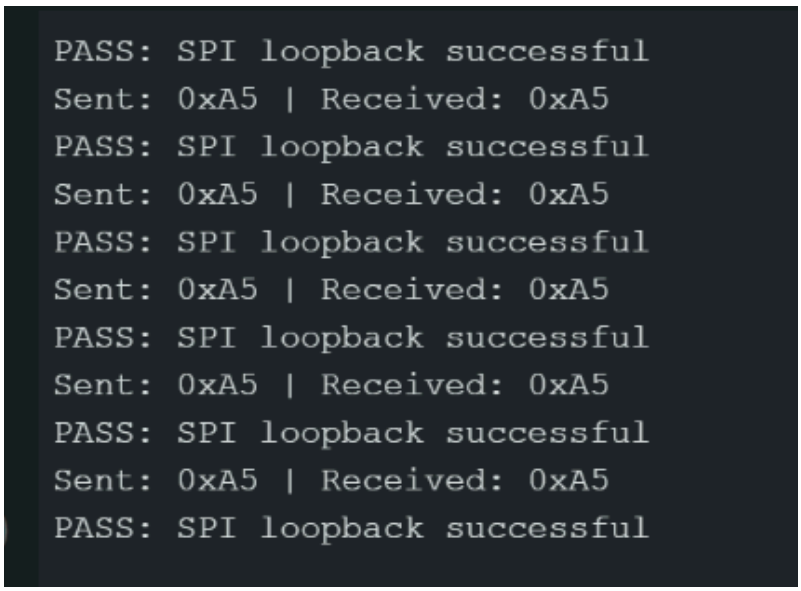
Figure 2.2. Failed I²C Scanner Test



Observation: The scanner did not detect any I²C devices after the OLED was disconnected.

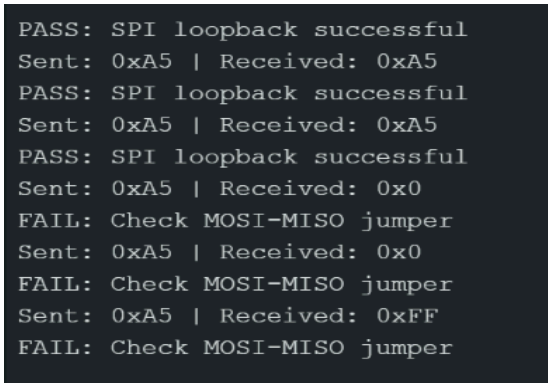
3. SPI Loopback Test

Figure 3.1. Successful SPI Loopback Test



Observation: The received data matched the transmitted data when MOSI was connected to MISO.

Figure 3.2. Failed SPI Loopback Test



Observation: The received data did not match the transmitted data after removing the MOSI–MISO jumper.